Inspection and Testing Plan
for
Piping Assembly Works

(Doc No. CT8-ITP-PI-001)

USA PROJECT UNIT 1 - TALLGRASS

REVISION HISTORY				
REVISION	**DATE**	**DESCRIPTIONS**	**PREPARED BY**	**APPROVED BY**
0	1 Apr 26	First Issue	Ms. Michelle Wilrutstien	Mr. Phisit Pew-La-Or

BJCHI		**BJCHI**		**CIMTAS**	
Prepared by (QA Engineer):	Ms. Michelle Wilrutstien	**Approved by (QAQC Manager):**	Mr. Phisit Pew-La-Or	**Approved by (Client):**	REVIEWED BY QC INSPECTOR SIGNATURE: DATE: CIMTAS NINGBO
Date:	01-Apr-26	**Date:**	01-Apr-26	**Date:**	Digitally signed by Suriya Lamor Date: 2026.04.02 11:06:55 +07'00'
Signature:		**Signature:**		**Signature:**	

Job/Project No.:	CT8	Project Name:	USA PROJECT UNIT 1 - TALLGRASS	Location:	☑ Shop ☐ Site

No.	Production and Inspection Items	Specifications and Procedures			Inspection & Test Equipment	Inspection Marking	Verifying Control Records	Inspection Points											
								BJC			CIMTAS			Mitsubishi Power			AI/NOBO		
		CODE	BJC / CIMTAS	Mitsubishi Power				IP	Date	Sign	IP	Date	Sign	IP	Date	Sign	IP	Date	Sign
1	**SPECIFICATIONS**																		
A	Verify that all relevant documents have been received	-	-	Project Specification	-	-	-	R			-			-			-		
2	**ENGINEERING**																		
A	Engineering Documents	-	Technical Specification	LU-00130 Latest rev.	-	-	-	R			-			-			-		
B	AFC Drawings	-	Drawings/ Part Lists		-	-	-	R			-			-			-		
3	**WELDING QUALITY CONTROL AND NDT**																		
A	WPS/PQR	ASME Sec IX, ASME B31.1	CT8-PRO-PI-001 (WPS/PQR List for Piping Work)	LU-00130 Latest rev.	-	-	Signed / Approved WPS & register	R			R			R			-		
B	Welder Qualification register	ASME Sec IX,	CT8-PRO-PI-002 (WQP List for Piping Work)		-	-		R			R			R			-		
4	**MATERIAL RECEIVING (PIPING)**																		
4.1	**MATERIAL SUPPLIED BY CLIENT**																		
A	Review Mat'l list / Certificate (if any)	-	CT8-PRO-CM-009 Material Handling & Storage Procedure / MRR & RIR/ AFC Dwg. & Packing List	LU-00130 Latest rev.	Visual Inspection & Measuring Tools	Review Stamp	MRR, RIR & OS&D Report	R			R			R			-		
B	Check condition of packing							R			R			-			-		
C	Check quantity of Material							H			W/R			-			-		
D	Check surface condition of Material (Inside & outside) Note: All unaccepted indications on Material shall be informed to client within 48 hours							H			W/R			R			-		
5	**MATERIAL RECEIVING (PIPE SUPPORTS)**																		
5.1	**DOCUMENT REVIEW BEFORE ORDERING**																		
A	Review MPS for Material requisition compliance Specification	ASME Sec II Part A, Relevant Design Code	CIMTAS Agreement, PO, Technical requirements	LU-00130 Latest rev.	-	-	-	R			-			-			-		
B	Review MPS for Welding consumable requisition compliance Specification				-	-	-	R			-			-			-		
5.2	**MATERIAL RECEIVING INSPECTION**																		
A	Check Packing, Material Physical, Quantity, Dimension	ASME Sec II Part A, Relevant Design Code	CT8-PRO-CM-009 Material Handling & Storage / MRR & RIR	LU-00130 Latest rev.	Visual Inspection & Measuring Tools	Accepted Stamp	Material Receiving Report	H			R			R			-		
B	Material & Mill cert. verification. Check Chemical & Mechanical for Compliance & requirement as per specification.					Reviewed Stamp	Material Certificate	R			R			R			-		
C	Check mark & designation					Reviewed Stamp	Material Certificate	H			M			R			-		

Legend: A: Approve H: Hold Point M: Monitoring Activity R: Document Review V: Verify W: Witness N/A: Not Applicable
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5.3	**Welding Consumable**																		
A	Check Packing, Material Physical, Quantity	ASME Sec II Part C	CT8-PRO-CM-002 – Welding Consumable Control Procedure and MRR / RIR	LU-00130 Latest rev.	Visual Inspection	Reviewed Stamp	Mat'l Certificate	H			R			R			-		
B	Welding consumable cert. verification							R			R			R			-		
5.4	**Paint Material**																		
A	Check Packing, Material Physical, Quantity	Manufacturer data sheet	CT8-PRO-CM-009 Material Handling and Storage Procedure / MRR & RIR	LU-00130 Latest rev.	Visual Inspection	Reviewed Stamp	Material Certificate & MSDS	H			R			R			-		
B	Painting consumable cert. verification							R			R			R			-		
5.5	**Insulation Material**																		
A	Check Packing, Material Surface, Quantity	Manufacturer data sheet	CT8-PRO-CM-009 Material Handling and Storage Procedure / MRR & RIR	LU-00130 Latest rev.	Visual Inspection	Reviewed Stamp	Material Certificate & MSDS	H			R			R			-		
B	Material spec, cert. verification							R			**R**			R			-		
6	**MATERIAL PREPARATION**																		
6.1	**Pipe Spool Preservation**																		
A	Ensure Spool/Valves are correctly tagged with relevant details	-	CT8-PRO-CM-010 – Packing, Tagging and Preservation Procedure	LU-00130 Latest rev.	Visual Inspection & Inspection Tools	-	Photo Report	H			M			**R**			-		
B	Ensure Spool / Valves are correctly cleaned, stored & protected during temporary storage							H			M			**R**			-		
7	**PIPING SPOOL ASSEMBLY**																		
A	Pipe Fit up & Pipe Spool Installation & Inspection	-	CT8-PRO-CM-004 Welding Control Procedure / CT8-PRO-PI-001 WPS-PQR Register / CT8-PRO-PI-002 Welder Qualified Register&Certificates / AFC Drawing	LU-00130 Latest rev.	Visual Inspection & Inspection Tools	Marked Evidence	Welding History Sheet	H			W			**W/R**			-		
B	Welding							H			**M**			**W/R**			-		
C	Visual and Dimensional Inspection							H			W			**W/R**			-		
8	**NON DESTRUCTIVE TEST (If require)**																		
A	100% Visual Inspection	ASME B31.1/ SNT-TC-1A	CT8-PRO-CM-012 - Visual Inspection Procedure	LU-00130 Latest rev.	Visual Inspection / MT / PT / RT Equipment	Marked Evident	VT / PT/ MT/ RT Report	H			R			R			-		
B	PT or MT as per project specification		CT8-PRO-PI-003 – PT Procedure for Piping / CT8-PRO-PI-004 – MT Procedure for Piping					W			R			R			-		
C	RT as per project specification		CT8-PRO-PI-005 – RT Procedure for Piping					W			R			R			-		

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9	**REPAIR AFTER NDT**																		
9.1	**Base metal repair (Edge preparation; Notch / Remove Temporary attachment / Mechanical damage)**																		
A	After weld repair confirm by MT or PT	ASME B31.1/ SNT-TC-1A	CT8-PRO-CM-003 Weld Repair Procedure/ CT8-PRO-CM-012 Visual Inspection Procedure/ CT8-PRO-PI-003 PT Procedure for Piping/ CT8-PRO-PI-004 MT Procedure for Piping	LU-00130 Latest rev.	Visual Inspection / MT / PT Equipment	-	MT or PT Report	H			R			R			-		
9.2	**Internal defect Repair**																		
A	Mark defective weld location	ASME B31.1/ SNT-TC-1A	CT8-PRO-CM-003 Weld Repair Procedure/ CT8-PRO-CM-012 Visual Inspection Procedure/ CT8-PRO-PI-003 PT Procedure for Piping/ CT8-PRO-PI-004 MT Procedure for Piping	LU-00130 Latest rev.	Visual Inspection / MT / PT Equipment	-	PT or MT Report	H			R			R			-		
B	Monitor Excavation Process							W			R			R			-		
C	MT after Excavation to confirm Removal of Defect							W			R			R			-		
9.3	**Post Repair Inspection**																		
A	Repeat Visual Inspection	ASME B31.1	CT8-PRO-CM-003 Weld Repair Procedure/ CT8-PRO-CM-012 Visual Inspection Procedure	LU-00130 Latest rev.	Visual Inspection	-	-	H			R			R			-		
9.4	**NDT (s) after completion of weld repair**																		
A	Repeat necessary NDT	ASME B31.1/ SNT-TC-1A	CT8-PRO-CM-003 Weld Repair Procedure/ CT8-PRO-CM-012 Visual Inspection Procedure/ CT8-PRO-PI-003 PT Procedure for Piping/ CT8-PRO-PI-004 MT Procedure for Piping / CT8-PRO-PI-005 RT Procedure for Piping	LU-00130 Latest rev.	Visual Inspection / PT / MT / RT Equipment	-	PT / MT or RT Report	W			R			R			-		
10	**EXCEPTIONS & SITE INSTRUCTIONS (If any)**																		
10.1	**Site Instruction**																		
A	Corrections on deviations from drawings or specifications	-	Drawing and Site Instruction or Deviation Documents	Drawing and Site Instruction or Deviation Documents	-	-	Site Instruction / Approved Deviation Documents	R			M			M			-		
B	Revise & approve engineering discrepancies found at site							R			M			M			-		
C	Proposal & approval for Mat'l substitution							R			M			M			-		
D	Modification as per Site Instruction							R			M			M			-		
11	**INSPECTION RELEASE NOTE (IRN)**																		
A	Issue IRN on System components reflecting all outstanding works (if any)	-	-	LU-00130 Latest rev.	-	-	IRN	R			R			R			-		

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12	**INSPECTION BEFORE PRESSURE TEST**																		
A	General Visual Inspection	ASME B31.1	CT8-PRO-CM-005 Dimension Control Procedure / AFC Drawing	LU-00130 Latest rev.	Visual Inspection & Total Station Camera	-	Final Dimension Inspection Report	H			W			W/R			-		
B	Overall Dimension Survey on Piping							H			W			W/R			-		
13	**PNEUMATIC TEST**																		
A	Line check prior to test	ASME B31.1Chapter VI para. 137.5	CT8-PRO-PI-007 – Pneumatic Testing Procedure (N2) / AFC Drawing	LU-00130 Latest rev.	Visual Inspection	-	Test Package & Pneumatic Testing Report	H			M			M			-		
B	Release for Pneumatic Test							R			R			R			-		
C	Perform Pneumatic test							H			W			W/R			-		
D	Internal Cleanliness of piping line		CT8-PRO-CM-014 – Module Assembly, Preservation & Cleanliness Procedure		Visual Inspection & Inspection Tools	-	Photo Report	H			W			W/R			-		
14	**RE-INSTALLATION AND OPEN END PROTECTION**																		
A	Check piping connection and slope	-	CT8-PRO-CM-008 - Bolt Tightening for Piping & Structure Procedure / AFC Drawing	LU-00130 Latest rev.	Visual Inspection, Measuring Tools, Torque Wrench	-	Torque Tightening Report / Line Check & Test Package Punch List	H			M			M			-		
B	Check pipe supports							H			M			M			-		
C	Check gasket type on flange connections							H			M			M			-		
D	Check bolts on flange connection tightened							H			M			M			-		
E	Check correct valves installed							H			M			M			-		
15	**TRIAL ASSEMBLY**																		
A	Assembly check and loose bolt	-	CT8-PRO-CM-005 – Dimension Control Procedure / AFC Drawing	LU-00130 Latest rev.	Visual Inspection, Measuring Tools & Total Station Camera. 3D Laser Tracking if required	-	Final Dimension Inspection Report / Loose Bolt Photo Report	H			W			W			-		
B	Dimension check of piping open end							H			W			W			-		
C	Modifications works (if required)							H			W			W			-		
16	**FINAL CLEANING AND TOUCH-UP PAINT AND PRESERVATION**																		
A	Dry Film Thickness Control	Manufacturer data sheets	CT8-PRO-CM-013 – Surface Preparation and Coating Inspection Procedure	LU-00130 Latest rev.	Visual Inspection & DFT Gauge	-	Painting Report	H			R			R			-		
B	Inside cleanliness and conservation control	ASME B31.1	CT8-PRO-CM-014 – Module Assembly, Preservation & Cleanliness Procedure		Visual Inspection & Inspection Tools (Borescope)	-	Photo Report	W			R			R			-		
17	**INSULATION INSPECTION AND REPORT**																		
A	Verify type and thickness of insulation material	-	CT8-PRO-CM-020 – Method Statement for Piping Hot Insulation	LU-00130 Latest rev.	Visual Inspection	Accepted Stamp and sign	Insulation Inspection Report	W			M			M			-		
B	Insulation Material, Insulation supports, Insulation Jacketing, Cladding, Fasteners and accessories installation	-						H			M			M			-		
C	Insulation Final Inspection	-						H			M			M			-		

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18	**TAGGING & PACKING**																		
A	Check Cleanliness	-	CT8-PRO-CM-010 Packing, Tagging & Preservation Procedure	LU-00130 Latest rev.	Visual Inspection	-	IRN	H			M			M			-		
B	Check the package sealing							H			M			M			-		
C	Check Tagging							H			M			M			-		
D	Check with Packing list							H			M			M			-		
E	Check actual package sizing dimensions							H			M			M			-		
19	**FINAL INSPECTION**																		
A	Inspection Before Delivery	-	-	LU-00130 Latest rev.	Visual Inspection	-	Certificate of Inspection (COI)	H			W			W			-		
B	Inspection Release Certificate							H			H			H			-		
20	**DOCUMENTATION**																		
A	Compile Final Certification Dossier	-	CT8-PRO-CM-006 e-MDR Procedure	LU-00130 Latest rev.		-	Electronic Manufacturer Data Record (e-MDR)	R			R			R			-		
B	Check and Compile Quality Records/Document							R			R			R			-		
C	Compile completion package and dossier for approval							R			R			R			-		
D	Modification as per Site Instruction							R			-			-			-		

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